

20/05

Resolución Guía 6

Parametrización y orientación de curvas

* Ejercicio 2 $\gamma = [0, 2] \rightarrow \mathbb{R}^2 / \vec{\gamma}(t) = (\cos(t), \sin(t))$

a) Demostrar rapidez constante

$$\|\gamma'(t)\| = \sqrt{\cos^2(t) + \sin^2(t)} = 1$$

b) Reparametrizar curva 4 veces más rápido

$$\gamma(\mu) = (\cos(4\mu), \sin(4\mu))$$

$$\gamma'(\mu) = (-4\sin(4\mu), 4\cos(4\mu))$$

$$\|\gamma'(\mu)\| = \sqrt{(4\cos(4\mu))^2 + (4\sin(4\mu))^2} = \sqrt{4^2} = 4 = 4 \cdot \|\gamma'(t)\|$$

acotar intervalos

$$\begin{aligned} 0 &\leq t \leq 2\pi \\ 0 &\leq \mu \leq \frac{\pi}{2} \end{aligned}$$

⇒ Hallar aceleración

$$\gamma''(\mu) = (-16\cos(4\mu), -16\sin(4\mu))$$

⇒ Hallar μ cuando pasa por $(0, 1) = \gamma(\mu) \rightarrow \mu = \pi/8$

$$\gamma'(\pi/8) = (-4, 0) \rightarrow \text{velocidad}$$

$$\gamma''(\pi/8) = (0, -16) \rightarrow \text{aceleración}$$

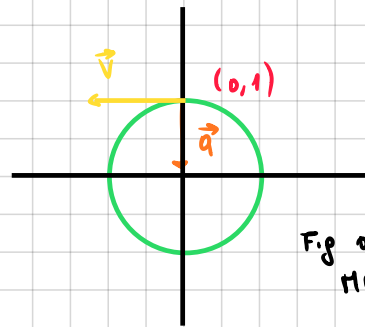


Fig análisis
MCU

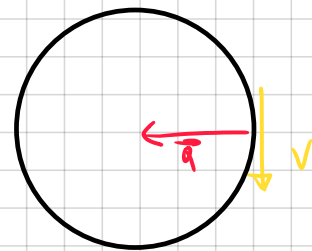
c) reparametrizar curva 2 veces más lento

$$\gamma(\mu) = (\cos(\frac{\mu}{2}), \sin(\frac{\mu}{2}))$$

$$\gamma'(\mu) = (-\frac{1}{2}\sin(\frac{\mu}{2}), \frac{1}{2}\cos(\frac{\mu}{2}))$$

$$\|\gamma'(\mu)\| = \sqrt{\left(\frac{1}{2}\right)^2 (\sin^2(\frac{\mu}{2}) + \cos^2(\frac{\mu}{2}))} = \frac{1}{2} \rightarrow \text{rapidez} = \frac{1}{2} \|\gamma'(t)\|$$

$\mu \in [0, 4\pi]$



Integral de campo escalares a lo largo de curvas

* Ejercicio 3: Calcular $\int_C f \, ds = \int_{t_0}^t f(p(t)) \|p'(t)\| \, dt$

a) $f(x,y) = 1/(x^2 + y^2)$ $C: x^2 + y^2 = 4, y \geq 0$ 1er y 2do cuadrante

↳ Hallar parametrización $C = \vec{p}(t) = (2\cos(t), 2\sin(t)), t \in [0, \pi]$

↳ Hallar $f(p(t))$

$$\bar{f}(\vec{p}(t)) = \frac{1}{(2\cos(t))^2 + (2\sin(t))^2} = \frac{1}{4}$$

↳ Hallar $\|p'(t)\|$

$$p'(t) = (-2\sin(t), 2\cos(t)) \Rightarrow \|p'(t)\| = \sqrt{(2\cos(t))^2 + (2\sin(t))^2} = 2$$

↳ $\oint_C f \, ds$

$$\oint_C f \, ds = \int_0^\pi \frac{1}{2} \, dt = \frac{\pi}{2}$$

b) $f(x,y,z) = 2x - yz$ C rect intersección π_1, π_2 / $\pi_1 = 2y - x + z = 2$, $\pi_2 = x - y + z = 4$ donde $a = (7, 4, 1)$ $b = (4, 2, 2)$

⇒ Hallar $r(t)$ intersección entre π_1, π_2

↳ $n_1 = (-1, 2, 1)$

↳ $n_2 = (1, -1, 1)$

$$\vec{v} = n_1 \times n_2 = \begin{vmatrix} i & j & k \\ -1 & 2 & 1 \\ 1 & -1 & 1 \end{vmatrix} = (3, 2, -1)$$

recta $r(t) = t(3, 2, -1) + (7, 4, 1) = (3t+7, 2t+4, -t+1)$

⇒ Hallar Intervalo de t

$$r(0) = 0 \cdot (3, 2, -1) + (7, 4, 1) = (7, 4, 1)$$

$$r(t) = t(3, 2, -1) + (7, 4, 1) = (4, 2, 2)$$

$$\begin{cases} 3t+7 = 4 \rightarrow 3t = -3 \\ 2t+4 = 2 \rightarrow 2t = -2 \\ -t+1 = 2 \rightarrow -t = 1 \end{cases} \rightarrow \boxed{t = -1}$$

⇒ Hallar $f(\vec{r}(t))$

$$\begin{aligned} f(\vec{r}(t)) &= 2(3t+7) - (2t+4)(-t+1) \\ &= (6t+14) - (-2t^2+2t-4t+4) \\ &= 6t+14 + 2t^2-2t+4t-4 \\ &= 2t^2+8t+10 \end{aligned}$$

⇒ Hallar $\|p'(t)\|$

$$p'(t) = (3, 2, -1) \quad \|p'(t)\| = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{14}$$

⇒ Hallar $\oint_C f \, ds$

$$\begin{aligned} \oint_C f \, ds &= \int_0^{-1} (2t^2+8t+10)\sqrt{14} \, dt = \sqrt{14} \left(2\int_0^{-1} t^2 \, dt + \int_0^{-1} 8t \, dt + \int_0^{-1} 10 \, dt \right) \\ &= \sqrt{14} \left(\frac{2}{3}t^3 + \frac{8}{2}t^2 + 10t \right) \\ &= \sqrt{14} \left(-\frac{2}{3} + 4 - 10 \right) = -\frac{20\sqrt{14}}{3} \end{aligned}$$

* Ejercicio 4 Calcular longitud

a) Curva $\vec{r}(t) = (t, \frac{2}{3}t^{\frac{3}{2}}, 2)$, $t \in [0, 3]$

$$\text{Long}(c) = \int_0^3 \|\sigma'(t)\| dt = \int_0^3 (1+t)^{\frac{1}{2}} dt = \frac{2}{3} (1+t)^{\frac{3}{2}} \Big|_0^3 = \frac{14}{3}$$

$$\sigma'(t) = (1, t^{\frac{1}{2}}, 0) = (1, \sqrt{t}, 0) \rightarrow \|\sigma'(t)\| = \sqrt{1+t} = \sqrt{1+t}$$

$$\hookrightarrow s(t) = \int_0^t \|\sigma'(\mu)\| d\mu = \frac{2}{3} (1+t)^{\frac{3}{2}} - \frac{2}{3}$$

b) Hélice $\vec{r} = (3\cos(t), 3\sin(t), 4t)$ $t \in [0, 2\pi]$

$$\text{Long}(c) = \int_0^{2\pi} \|\rho'(t)\| dt = \int_0^{2\pi} 5 dt = 10\pi$$

$$\vec{\rho}'(t) = (-3\sin(t), 3\cos(t), 4) \quad \|\rho'(t)\| = \sqrt{3^2(\sin^2 t + \cos^2 t) + 4^2} = \sqrt{25} = 5$$

$$\Rightarrow \text{reparam } s(t) = \int_0^t \|\rho'(\mu)\| d\mu = 5\mu \Big|_0^t = 5t \rightarrow s(2\pi) = 10\pi$$

* Ejercicio 5 Intersección $\Sigma_1: z = 2 - x^2 - 2y^2$ $\Sigma_2: z = x^2$

$\Rightarrow$ Hallar curva de intersección

$$\begin{cases} z = 2 - x^2 - 2y^2 \\ z = x^2 \end{cases} \rightarrow \begin{cases} z = 2 - z - 2y^2 \\ 2z = 2 - 2y^2 \end{cases} \rightarrow \begin{cases} z = 1 - y^2 \\ z = x^2 \end{cases} \rightarrow \begin{cases} x^2 = 1 - y^2 \\ y^2 + x^2 = 1 \end{cases} \rightarrow \text{curva } (\cos(t), \sin(t))$$

$\rightarrow$ Hallar $\|\rho'(t)\|$

$$\rho'(t) = \left(2 \frac{1}{\sqrt{1-t^2}} (-2t), 1, -2t \right)$$

$$\|\rho'(t)\| = \sqrt{\frac{t^2}{(1-t^2)} + 1 + 4t^2} = \sqrt{\frac{1}{(1-t^2)} + 4t^2}$$

$\rightarrow$ Hallar $\delta(\rho(t))$

$$\delta(\rho(t)) = k(\sqrt{1-t^2}, t)$$

$\rightarrow$ Hallar masa

$$M = \int_0^1 \delta(\rho(t)) \|\rho'(t)\| dt$$

$$\int k(\sqrt{1-t^2}, t) \sqrt{\frac{1}{(1-t^2)} + 4t^2} dt$$

$$= k \int t \cdot (\sqrt{1-t^2}) \frac{1}{\sqrt{1-t^2}} \cdot \sqrt{1+4t(1-t^2)} dt \rightarrow k \int t \sqrt{1+4t(1-t^2)} dt$$

$$\int \mu'v = \mu v - \int v\mu$$

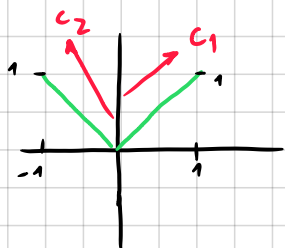
$$\int (\mu, v)' = \int \mu'v + \int v'\mu$$

$$(\mu v)' = \mu'v + v'\mu$$

(8)

* Ejercicio 6 : Hallar masa alambre V en $\mathbb{R}$ $y = |x|$ entre $(-1,1)$ y $(1,1)$, densidad $\delta = k|xy|$

⇒ Figura de análisis



Masa total = masa alambre $x \geq 0$
masa + alambre $x < 0$

$$m_1 + m_2 = \frac{2\sqrt{2}}{3} k$$

⇒ para $x \geq 0$ $C_1: \vec{p}_1(t) = (t, t)$

$$\left. \begin{array}{l} \hookrightarrow \vec{p}'_1(t) = (1, 1) \rightarrow \|\vec{p}'_1(t)\| = \sqrt{2} \\ \hookrightarrow \delta(\vec{p}_1(t)) = |t^2| \end{array} \right\} m_1 = \int_0^1 t^2 \cdot \sqrt{2} dt = \left. \frac{\sqrt{2} t^3}{3} \right|_0^1 = \frac{\sqrt{2}}{3} k$$

⇒ para $x < 0$ $C_2: \vec{p}_2(t) = (-t, t)$

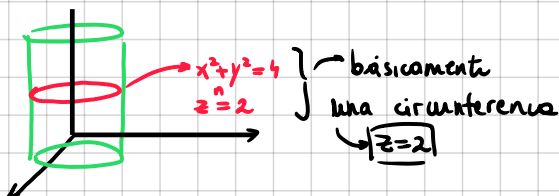
$$\left. \begin{array}{l} \hookrightarrow \vec{p}'_2(t) = (-1, 1) \rightarrow \|\vec{p}'_2(t)\| = \sqrt{2} \\ \hookrightarrow \delta(\vec{p}_2(t)) = |-t^2| = t^2 \end{array} \right\} m_2 = \int_{-1}^0 t^2 \sqrt{2} dt = \left. \frac{\sqrt{2} t^3}{3} \right|_{-1}^0 = \frac{\sqrt{2} (-1)^3}{3} = -\frac{\sqrt{2}}{3} k$$

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* Ejercicio 7 =

dist $x=0, y=0, z=0$

⇒ forma de alambre



⇒ Densidad en cada punto (x, y, z) o (x, y, z) ?

↳ "proporcional al producto de las distancias al plano coord"

$$\delta(x, y, z) = k \cdot |x| |y| |z|$$

⇒ parametrizar curva $\vec{g}(t) = (2\cos(t), 2\sin(t), 2), t \in [0, 2\pi]$

$$\hookrightarrow \vec{g}'(t) = (-2\sin(t), 2\cos(t), 0) \quad \|\vec{g}'(t)\| = \sqrt{(2\cos t)^2 + (2\sin t)^2} = 2$$

⇒ Hallar $\delta(\vec{g}(t))$

$$\delta(\vec{g}(t)) = k |2\cos(t)| |2\sin(t)| 2 = 8k |\cos(t)\sin(t)| \quad M = \int_0^{2\pi} 16k |\cos(t)\sin(t)| dt$$

$$= 8k \int_0^{2\pi} \sin(2t) dt = \left. \frac{8k (-\cos(2t))}{2} \right|_0^{2\pi} = -4k \cos(2t) \Big|_0^{2\pi} = -4k \cos(4\pi) - (-4k \cos(0)) =$$

? WTF

*Ejercicio 8 Hélice: $\vec{X} = \vec{g}(t) = (\cos(t), \sin(t), t), t \in [0, 2\pi]$ $\delta(x, y, z) = k(x^2 + y^2 + z^2)$

Paso:

- hallar masa total del alambre
- hallar densidad media
- hallar centro de masa $\rightarrow$ las masas por los dos lados de la curva equivalentes

$\Rightarrow$ Hallar $\|\vec{g}'(t)\|$

$$\vec{g}'(t) = (-\sin(t), \cos(t), 1) \Rightarrow \|\vec{g}'(t)\| = \sqrt{\cos^2 + \sin^2 + 1} = \sqrt{2}$$

$\Rightarrow$ Hallar $\delta(\vec{g}(t))$

$$\delta(\vec{g}(t)) = k(\cos^2(t) + \sin^2(t) + t^2) = k(1+t^2)$$

$\Rightarrow$ Hallar Masa total $\rightarrow M = \int_0^{2\pi} k(1+t^2) \sqrt{2} dt = \sqrt{2} k \int_0^{2\pi} (1+t^2) dt = \sqrt{2} k \left(\int_0^{2\pi} 1 dt + \int_0^{2\pi} t^2 dt \right)$

$$M = \sqrt{2} k \left(t \Big|_0^{2\pi} - \frac{t^3}{3} \Big|_0^{2\pi} \right) = 12k \left(2\pi - \frac{(2\pi)^3}{3} \right) = 12k \left(\frac{6\pi - (2\pi)^3}{3} \right)$$

$$M = 4k(6\pi - (2\pi)^3) \xrightarrow{\text{tempo Down}} \sqrt{2} k \left(2\pi + \frac{(2\pi)^3}{3} \right)$$

$\Rightarrow$ Hallar longitud alambre

$$\text{longitud alambre} = \int_0^{2\pi} \|\vec{g}'(t)\| dt = \int_0^{2\pi} \sqrt{2} dt = \sqrt{2} t \Big|_0^{2\pi} = \sqrt{2} \cdot 2\pi$$

$\Rightarrow$ Hallar densidad media

$$\delta_{\text{media}} = M_{\text{total}} / \text{long} = \frac{4k(6\pi - (2\pi)^3)}{\sqrt{2} \cdot 2\pi} = \frac{2k(6 - 8\pi^2)}{\sqrt{2}}$$

$\Rightarrow$ Hallar centro de masa

m_2 C_m m_1

$\rightarrow$ función $m_1(x) = \int_0^x \delta(\vec{g}(t)) \cdot \|\vec{g}'(t)\| dt = 12\sqrt{k} \left(t \Big|_0^x - \frac{t^3}{3} \Big|_0^x \right)$

$$m_1(x) = 12\sqrt{k} \left(x - \frac{x^3}{3} \right)$$

Alambre

$\rightarrow$ Aquí estarán buscando pts med.

no es lo mismo que centro de masa

$$m_2(x) = \int_x^{2\pi} \delta(\vec{g}(t)) \cdot \|\vec{g}'(t)\| dt = 12\sqrt{k} \left(t \Big|_x^{2\pi} - \frac{t^3}{3} \Big|_x^{2\pi} \right)$$

$$m_2(x) = 12\sqrt{k} \left(2\pi - x - \left(\frac{(2\pi)^3}{3} - \frac{x^3}{3} \right) \right)$$

Hallar x para que $m_1(x) = m_2(x)$

$$12\sqrt{k} \left(x - \frac{x^3}{3} \right) = 12\sqrt{k} \left(2\pi - x - \frac{(2\pi)^3}{3} + \frac{x^3}{3} \right)$$

$$x - \frac{x^3}{3} = 2\pi - x - \frac{(2\pi)^3}{3} + \frac{x^3}{3}$$

$$2x - \frac{2x^3}{3} = 2\pi - \frac{(2\pi)^3}{3} \rightarrow \boxed{x = \pi}$$

$\rightarrow$ punto $\vec{g}(\pi) = (1, 0, \pi)$
centro de masa

(Rehacer ejercicio 8)

*Ejercicio 8 Hélice: $\vec{r} = \vec{g}(t) = (\cos(t), \sin(t), t), t \in [0, 2\pi]$

$$\delta(x, y, z) = k(x^2 + y^2 + z^2)$$

Paso 1:

- hallar masa total del alambre
- hallar densidad media
- hallar centro de masa $\rightarrow$ las masas por los dos lados de la curva equivalentes

$\Rightarrow$ Hallar $\|\vec{g}'(t)\|$

$$\vec{g}'(t) = (-\sin(t), \cos(t), 1) \Rightarrow \|\vec{g}'(t)\| = \sqrt{\cos^2 + \sin^2 + 1} = \sqrt{2}$$

$\Rightarrow$ Hallar $\delta(\vec{g}(t))$

$$\delta(\vec{g}(t)) = k(\cos^2(t) + \sin^2(t) + t^2) = k(1 + t^2)$$

$$\Rightarrow \text{Hallar Masa total} \rightarrow M = \int_0^{2\pi} k(1+t^2) \sqrt{2} dt = \sqrt{2} k \int_0^{2\pi} (1+t^2) dt = \sqrt{2} k \left(\int_0^{2\pi} 1 dt + \int_0^{2\pi} t^2 dt \right)$$

$$M = \sqrt{2} k \left(t \Big|_0^{2\pi} + \frac{t^3}{3} \Big|_0^{2\pi} \right) = \sqrt{2} k \left(2\pi + \frac{(2\pi)^3}{3} \right)$$

$$M = \sqrt{2} k \left(2\pi + \frac{(2\pi)^3}{3} \right)$$

$\Rightarrow$ Hallar longitud alambre

$$\text{longitud alambre} = \int_0^{2\pi} \|\vec{g}'(t)\| dt = \int_0^{2\pi} \sqrt{2} dt = \sqrt{2} t \Big|_0^{2\pi} = \sqrt{2} \cdot 2\pi$$

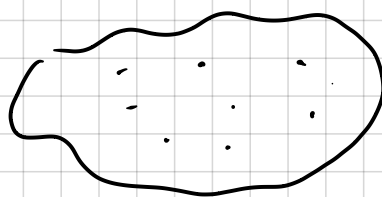
$\Rightarrow$ Hallar densidad media

$$\delta_{\text{media}} = M_{\text{total}} / \text{long} = \frac{\sqrt{2} k (2\pi + (8\pi^3)/3)}{\sqrt{2} \cdot 2\pi} = \underbrace{k \left(1 + \frac{4\pi^2}{3} \right)}_{\text{masa media}}$$

$\Rightarrow$ Hallar centro de masa

$$\vec{r}(t) = (x(t), y(t), z(t)) \quad M = \int_C \delta ds$$

$$\vec{r}_{\text{cm}} = \frac{1}{M} \int_C \vec{r} \cdot \delta ds$$



$$CM = \frac{\sum \text{masa elemento} \times \text{posicion}}{\text{masa total}}$$

$$\vec{R}_{\text{cm}} = \frac{\sum m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_n \vec{r}_n}{m_{\text{total}}}$$

$\rightarrow$ Hallar x_{pos}

$$x_{\text{pos}} = \frac{1}{M} \int_0^{2\pi} \cos(t) \delta(\vec{g}(t)) \|\vec{g}'(t)\| dt = \frac{\sqrt{2} k}{M} \int_0^{2\pi} \cos(t) (1+t^2) dt = \frac{\sqrt{2} k}{M} \left(\int_0^{2\pi} \cos(t) dt + \int_0^{2\pi} t^2 \cos(t) dt \right)$$

$$x_{\text{cm}} = \frac{\sqrt{2} k}{M} - \frac{2\pi}{M}$$

que voyon

$$\begin{aligned} \mu &= (1+t^2) \rightarrow \mu' = (2t) & \mu &= 2t \rightarrow \mu' = 2 \\ v &= \cos(t) \rightarrow v = -\sin(t) & v &= \sin(t) \rightarrow v = \cos(t) \end{aligned}$$

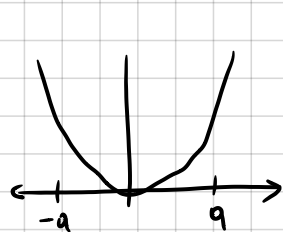
$$= \frac{\sqrt{2} k}{M} \left(\sin(t) (1+t^2) - \left(2t \cos(t) + 2 \sin(t) \right) \Big|_0^{2\pi} \right)$$

$$= \frac{\sqrt{2} k}{M} \left(\sin(2\pi) (1+(2\pi)^2) - (2\pi \cos(2\pi) + 2 \sin(2\pi)) \right)$$

$$= -2\pi$$

* Ejercicio 9 Inercia $\int r^2 \cdot dm = \int r^2 \delta ds = \int r^2 \delta(\vec{r}(t)) \|\vec{r}'(t)\| dt$

↳ alambre $\vec{r}(t) = [-a, a] \subset \mathbb{R} \rightarrow \mathbb{R}^2 / \vec{r}(t) = (t, \cosh(t))$



dist eje $y = |x|$
dist eje $x = |y|$

$$I_x = \int_{-a}^a |y|^2 \delta(\vec{r}(t)) \|\vec{r}'(t)\| dt$$

$$I_y = \int_{-a}^a |x|^2 \delta(\vec{r}(t)) \|\vec{r}'(t)\| dt$$

⇒ Hallar $\|\vec{r}'(t)\|$

$$\vec{r}'(t) = (1, \sinh t)$$

$$\|\vec{r}'(t)\| = \sqrt{1 + \sinh^2 t} = \sqrt{\cosh^2 t}$$

$$\cosh^2 t = \sinh^2 t + 1$$

$$\boxed{\cosh^2 t - \sinh^2 t = 1}$$

⇒ Hallar $\delta(\vec{r}(t)) = \delta_0$

$$I_x = \int_{-a}^a \delta_0 \cdot (\cosh t)^2 \sqrt{1 + \sinh^2 t} dt = \delta_0 \int_{-a}^a (\cosh t)^2 \sqrt{\cosh^2 t} dt = \delta_0 \int_{-a}^a \cosh^3 t dt$$

$$\delta_0 \int_{-a}^a \cosh^3 t dt = \delta_0 \int_{-a}^a (\sinh^2 t + 1) \cosh t dt = \delta_0 \int_{-a}^a \mu d\mu = \delta_0 \frac{\mu^2}{2} \Big|_{-a}^a = \delta_0 \cdot \frac{(\sinh^2 t + 1)^2}{2} \Big|_{-a}^a$$

$\mu = \sinh^2 t + 1$
 $d\mu = 2 \sinh t dt$

$$I_y = \int_{-a}^a \delta_0 t^2 \cosh t dt = -t^2 \sinh t - 2t \cosh t + 2 \sinh t \Big|_{-a}^a$$

$$\int \delta_0 t^2 \cosh t dt = t^2 \sinh t + \int 2t \sinh t dt = t^2 \sinh t + 2t \cosh t - 2 \sinh t$$

$$\mu = t^2 \rightarrow \mu' = 2t$$

$$v' = \cosh t \Leftrightarrow v = \sinh t$$

$$\mu = 2t \rightarrow \mu' = 2$$

$$v' = \sinh t \rightarrow v = \cosh t$$

* Ejercicio 11

$$C: \lambda \in [a, b] \subset \mathbb{R} \rightarrow \mathbb{R}^2 \quad \lambda \in C^1 \quad \text{len}(C) = 4 \quad \left. \begin{array}{l} \text{long } C = \int_a^b \|\lambda'\| dt = 4 \\ f(\lambda) = 3 \end{array} \right\}$$

$$\text{es } N_3 \text{ de } f: U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$$

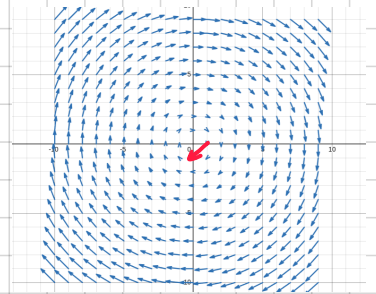
$$\int_C f ds = \int_a^b f(\lambda) \cdot \|\lambda'\| dt = 3 \int_a^b \|\lambda'\| dt = 3 \cdot 4 = 12$$

Valor medio de f es 3 ya que es constante en C

↳ porque C es N_3

Integral de campos vectoriales (Circulación)

* Ejercicio 12 = Circulación de $\vec{f}(x,y) = (y, -x)$ (1,0) a (0,-1)

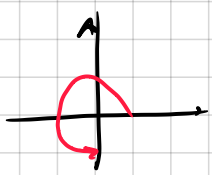


a) Segmento que los une $\vec{\lambda}(t) = t(1,1) + (1,0) = (t+1, t)$

$$\int_C \vec{f} \cdot d\vec{s} = \int_0^{-1} \vec{f}(\vec{\lambda}(t)) \cdot \vec{\lambda}'(t) dt = \int_0^{-1} (t, -t-1) (1,1) dt = \int_0^{-1} t - t - 1 dt = -t \Big|_0^{-1} = 1$$

$$\vec{f}(\vec{\lambda}(t)) = (t, -t-1) \quad \vec{\lambda}'(t) = (1,1)$$

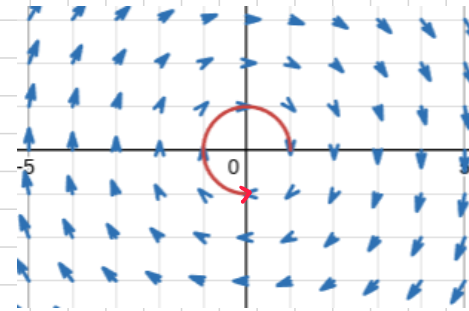
b) 3/4 circunferencia radio 1



$$\vec{\rho}(t) = (\cos(t), \sin(t)), t \in [0, \frac{3\pi}{2}]$$

$$\vec{f}(\vec{\rho}(t)) = (\sin(t), -\cos(t)) \quad \vec{\rho}'(t) = (-\sin(t), \cos(t))$$

$$\begin{aligned} \int_C \vec{f} \cdot d\vec{s} &= \int_0^{\frac{3\pi}{2}} (\sin(t), -\cos(t)) \cdot (-\sin(t), \cos(t)) dt \\ &= \int_0^{\frac{3\pi}{2}} -\sin^2 t - \cos^2 t dt = - \int_0^{\frac{3\pi}{2}} 1 dt = -t \Big|_0^{\frac{3\pi}{2}} = -\frac{3\pi}{2} \end{aligned}$$

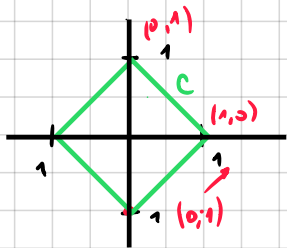
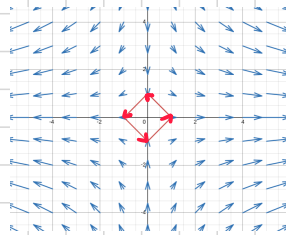


* Ejercicio 13

$$\vec{f}(x,y) = (2x, -y)$$

a) $\int_{C^+} (2x, -y) d\vec{s} \quad C: |x| + |y| = 1$

$$C = C_1 + C_2 + C_3 + C_4$$



$$\int_{C^+} \vec{f} \cdot d\vec{s} = \int_{C_1^+} \vec{f} \cdot d\vec{s} + \int_{C_2^+} \vec{f} \cdot d\vec{s} + \int_{C_3^+} \vec{f} \cdot d\vec{s} + \int_{C_4^+} \vec{f} \cdot d\vec{s} = \left(2 \cdot \frac{3}{2}\right) + 2 \cdot \left(-\frac{3}{2}\right) = 0$$

$$C_1: \vec{\rho}_1(t) = (t, t-1) \quad \vec{\rho}_1'(t) = (1,1) \quad \vec{f}(\vec{\rho}_1(t)) = (2t, -t+1)$$

$$\int_{C_1^+} \vec{f} \cdot d\vec{s} = \int_0^1 (2t, -t+1) \cdot (1,1) dt = \int_0^1 2t + 1 - t dt = \int_0^1 t + 1 dt = \frac{t^2}{2} + t \Big|_0^1 = \frac{1}{2} + 1 = \frac{3}{2}$$

$$C_2: \vec{\rho}_2(t) = (-t+1, t) \quad \vec{\rho}_2'(t) = (-1,1) \quad \vec{f}(\vec{\rho}_2(t)) = (-2t+2, -t)$$

$$\int_{C_2^+} \vec{f} \cdot d\vec{s} = \int_0^1 (-2t+2, -t) \cdot (-1,1) dt = \int_0^1 2t - 2 - t dt = \int_0^1 t - 2 dt = \frac{t^2}{2} - 2t \Big|_0^1 = \frac{1}{2} - 2 = -\frac{3}{2}$$

$$C_3: \vec{\rho}_3(t) = (-t, -t+1) \quad \vec{\rho}_3'(t) = (-1,-1) \quad \vec{f}(\vec{\rho}_3(t)) = (-2t, t-1)$$

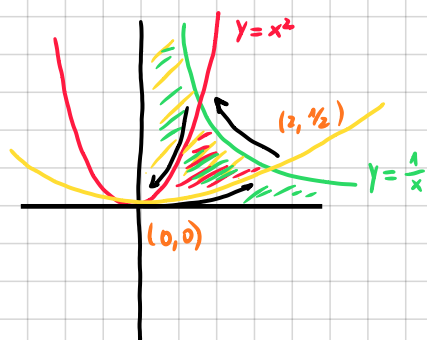
$$\int_{C_3^+} \vec{f} \cdot d\vec{s} = \int_0^1 (-2t, t-1) \cdot (-1,-1) dt = \int_0^1 2t - t + 1 dt = \int_0^1 t + 1 dt = \frac{t^2}{2} + t \Big|_0^1 = \frac{1}{2} + 1 = \frac{3}{2}$$

$$C_4: \vec{\rho}_4(t) = (t-1, -t) \quad \vec{\rho}_4'(t) = (1,-1) \quad \vec{f}(\vec{\rho}_4(t)) = (2t-2, -t)$$

$$\int_{C_4^+} \vec{f} \cdot d\vec{s} = \int_0^1 (2t-2, -t) \cdot (1,-1) dt = \int_0^1 2t - 2 - t dt = \int_0^1 t - 2 dt = \frac{t^2}{2} - 2t \Big|_0^1 = \frac{1}{2} - 2 = -\frac{3}{2}$$

⑥ $\int_{C^+} (xy, x^2) d\vec{s}$ C es frontera región primer cuadrante $xy \leq 1, y \leq x^2, y \geq x^2$

$\Rightarrow$ Hallar intersección $xy \leq 1, y \leq x^2, y \geq x^2$ $f(x,y) = (xy, x^2)$



Hallar punto de intersección

$$y = \frac{1}{x} \quad y = x^2$$

$$\frac{1}{x} = \frac{1}{8} x^2 \rightarrow x^3 = 8 \rightarrow \boxed{x=2}$$

punto $(2, \frac{1}{2})$

$$y = \frac{1}{x} \quad y = x^2$$

$$x^2 = \frac{1}{x} \Rightarrow x^3 = 1 \rightarrow \boxed{x=1}$$

punto $(1, 1)$

$$C = C_1 + C_2 + C_3$$

$$C_1: \vec{g}_1(t) = (t, \frac{1}{8}t^3), t \in [0, 2] \quad \vec{g}_1'(t) = (1, \frac{1}{4}t^2) \quad f(\vec{g}_1(t)) = (\frac{1}{8}t^3, t^2)$$

$$C_2: \vec{g}_2(t) = (t, 1/t), t \in [2, 1] \quad \vec{g}_2'(t) = (1, -1/t^2) \quad f(\vec{g}_2(t)) = (1, t^2)$$

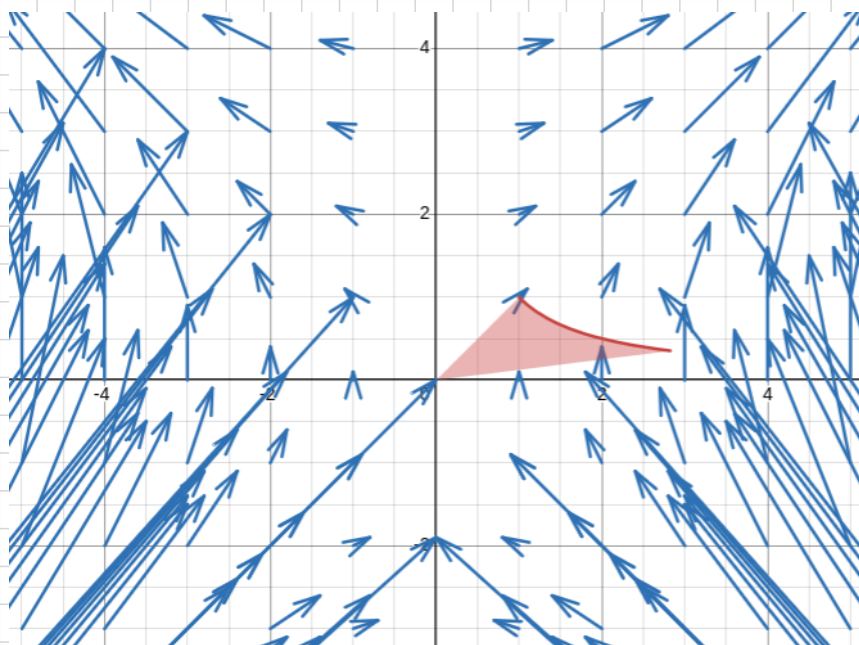
$$C_3: \vec{g}_3(t) = (t, t^2), t \in [1, 0] \quad \vec{g}_3'(t) = (1, 2t) \quad f(\vec{g}_3(t)) = (t^3, t^2)$$

$$\int_{C_1^+} \vec{f} d\vec{s} = \int_0^2 (\frac{1}{8}t^3, t^2) (1, \frac{1}{4}t^2) dt = \int_0^2 \frac{1}{8}t^3 + \frac{1}{4}t^3 dt = \frac{3}{8} \frac{t^4}{4} \Big|_0^2 = \frac{3}{2}$$

$$\int_{C_2^+} \vec{f} d\vec{s} = \int_2^1 (1, t^2) (1, -1/t^2) dt = \int_2^1 1 - 1 dt = \int_2^1 0 dt = 0$$

$$\int_{C_3^+} \vec{f} d\vec{s} = \int_1^0 (t^3, t^2) (1, 2t) dt = \int_1^0 t^3 + 2t^3 dt = \frac{3}{4} \frac{t^4}{4} \Big|_1^0 = -\frac{3}{4}$$

$$\int_{C^+} \vec{f} d\vec{s} = \int_{C_1^+} \vec{f} d\vec{s} + \int_{C_2^+} \vec{f} d\vec{s} + \int_{C_3^+} \vec{f} d\vec{s} = \frac{3}{2} + 0 + \frac{3}{4} = \frac{3}{4}$$

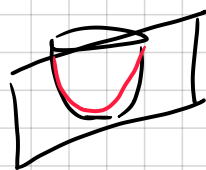


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* Ejercicio 14

(a) $\vec{\gamma}(t) = (t, t^2, 2t)$ $\vec{f}(x, y, z) = (xy, x, zy)$

$\Rightarrow C$ es una parábola $\rightarrow$ probablemente intersección de paraboloides con un plano

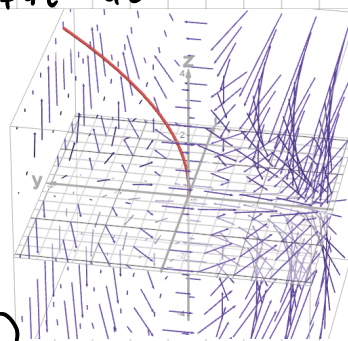


$\Rightarrow$ Hallar datos necesarios

$$\vec{f}(\vec{\gamma}(t)) = (t^3, t, 2t^3) \quad \vec{\gamma}'(t) = (1, 2t, 2)$$

$$\begin{aligned} \int_C \vec{f} d\vec{s} &= \int_0^2 \vec{f}(\vec{\gamma}(t)) \vec{\gamma}'(t) dt = \int_0^2 (t^3, t, 2t^3) (1, 2t, 2) dt = \int_0^2 (t^3 + 2t^4 + 4t^3) dt \\ &= \int_0^2 (5t^3 + 2t^2) dt = \left[\frac{5}{4}t^4 + \frac{2}{3}t^3 \right]_0^2 = \frac{76}{3} \end{aligned}$$

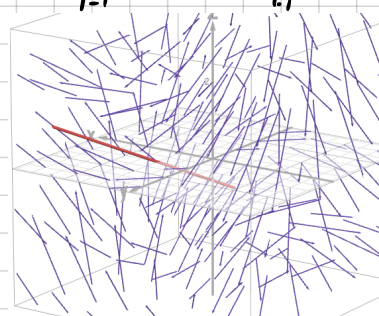
punto inicial es $\vec{\gamma}(0) = (0, 0, 0)$ y $\vec{\gamma}(2) = (2, 4, 4)$



(b) $\vec{\gamma}(t) = (t+1, 2t+1, t)$ $t \in [-1, 2]$ $\vec{f}(x, y, z) = (x+2y+z, 2y, 3x-z)$

$$\vec{f}(x, y, z) = ((t+1)+2(2t+1)+t, 2(2t+1), 3(t+1)-t) \quad \vec{\gamma}'(t) = (1, 2, 1)$$

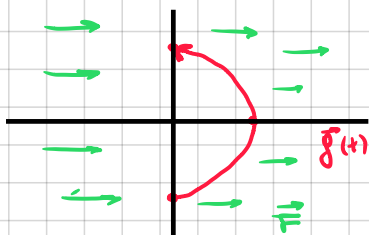
$$\begin{aligned} \int_C \vec{f} d\vec{s} &= \int_{-1}^2 (6t+3, 4t+2, 2t+3) (1, 2, 1) dt = \int_{-1}^2 (6t+3+8t+4+2t+3) dt \\ &= \int_{-1}^2 (16t+10) dt = \left[\frac{16}{2}t^2 + 10t \right]_{-1}^2 = 8t^2 + 10t \Big|_{-1}^2 = 52 - (-2) = 54 \end{aligned}$$



* Ejercicio 15 $\vec{F}(x, y, z) = 2 \cdot \vec{j} = (0, 2)$

$\Rightarrow$ f.p. análisis curva + cam. vel

W_F



$\Rightarrow$ parametrizar circunferencia $(0, -1) \rightarrow (1, 0) \rightarrow (0, 1)$

$$\vec{\gamma}(t) = (\sin(t), -\cos(t)), \quad t \in [0, \pi]$$

$$\vec{\gamma}(0) = (0, -1), \quad \vec{\gamma}(\frac{\pi}{2}) = (1, 0), \quad \vec{\gamma}(\pi) = (0, 1)$$

$$\vec{F}(\vec{\gamma}(t)) = (0, 2) \quad \vec{\gamma}'(t) = (\cos(t), \sin(t)) \quad t \in [0, \pi]$$

$$W_F = \int_0^\pi (0, 2) (\cos(t), \sin(t)) dt = 2 \int_0^\pi \sin(t) dt = -2 \cos(t) \Big|_0^\pi = \left(\underbrace{-2 \cos(\pi)}_2 \right) - \left(\underbrace{-2 \cos(0)}_{-2} \right) = 4$$

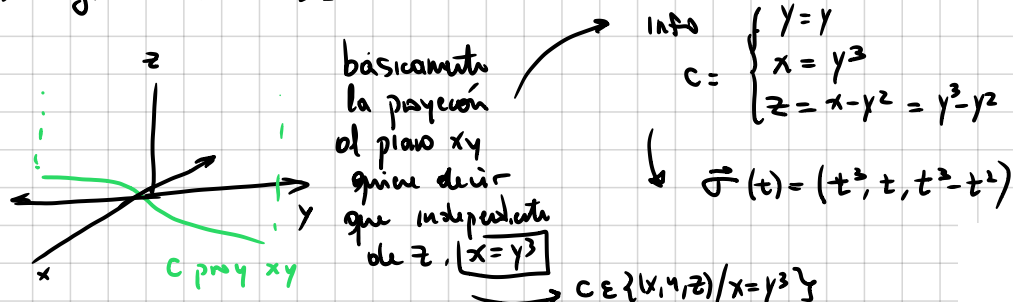
* Ejercicio 1b

$$\vec{F}(x,y,z) = (2g(x,y,z), xy - 9xg(x,y,z), 3yg(x,y,z)) \quad , \quad g: \mathbb{R}^3 \rightarrow \mathbb{R}$$

→ Calc circulación de $\vec{F}$ desde $(1, y_0, z_0)$ a $(8, y_1, z_1)$ $C \in \{z = x - y^2\}$ y C proy xy , $x = y^3$

→ Hallar C , parametrizar C , hallar punto inicial/final, calcular circulación

⇒ Figura de análisis C



⇒ Hallar valor inicial y final de t $t \in [1, 2]$

↳ hallar t_0 / $\vec{\sigma}(t_0) = (1, y_0, z_0)$

↳ Hallar t_1 / $\vec{\sigma}(t_1) = (8, y_1, z_1)$

$$\begin{cases} t_0^3 = 1 \rightarrow \boxed{t_0 = 1} \\ t_0 = y_0 \\ t_0^3 - t_0^2 = z_0 \end{cases}$$

$$\begin{cases} t_1^3 = 8 \rightarrow \boxed{t_1 = 2} \\ t_1 = y_1 \\ t_1^3 - t_1^2 = z_1 \end{cases}$$

⇒ Hallar datos necesarios para calcular circulación

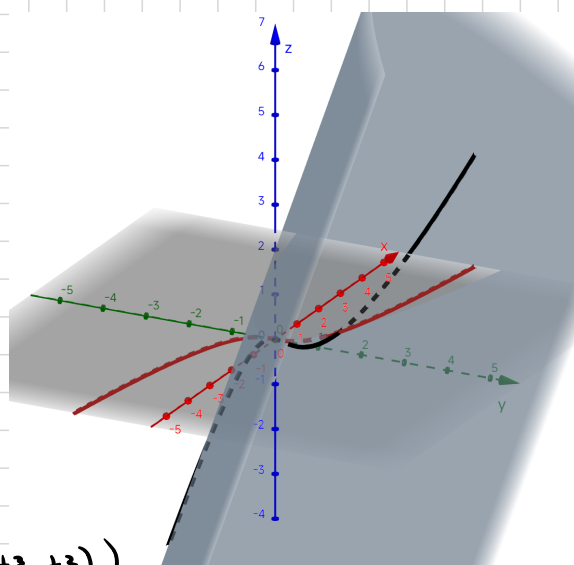
$$\vec{F}(\vec{\sigma}(t)) = (2 \cdot \vec{g}(t^3, t, t^3-t^2), t^4 - 9t^2 \vec{g}(t^3, t, t^3-t^2), 3t \vec{g}(t^3, t, t^3-t^2))$$

$$\vec{\sigma}'(t) = (3t^2, 1, 3t^2 - 2t)$$

⇒ Calcular circulación

$$\begin{aligned} \int_1^2 \vec{F}(\vec{\sigma}(t)) \cdot \vec{\sigma}'(t) dt &= \int_1^2 (2 \cdot \vec{g}(t^3, t, t^3-t^2), t^4 - 9t^2 \vec{g}(t^3, t, t^3-t^2), 3t \vec{g}(t^3, t, t^3-t^2)) \cdot (3t^2, 1, 3t^2 - 2t) dt \\ &= \int_1^2 (2\vec{g}, t^4 - 9t^2 \vec{g}, 3t \vec{g}) \cdot (3t^2, 1, 3t^2 - 2t) dt = \int_1^2 (6t^2 \vec{g} + t^4 - 9t^2 \vec{g} + 9t^2 \vec{g} - 6t^2 \vec{g}) dt \\ &= \int_1^2 t^4 dt = \frac{t^5}{5} \Big|_1^2 = \frac{32}{5} - \frac{1}{5} = \frac{31}{5} \end{aligned}$$

↳ El resultado no depende de $\vec{g}$



* Ejercicio 17

$$\vec{f}(x, y, z) = (x + \vec{g}(xy+z), y + \vec{g}(xy+z), 2z) \quad \vec{g}: \mathbb{R}^3 \rightarrow \mathbb{R} / \vec{g}(x, y, z) = xy + z$$

$$A = (2, 0, 3) \quad B = (3, -1, 6) \quad \text{dando } \vec{F}_{AB} \text{ circulación no dep. } \vec{g}$$

→ Hallar curva que pase por seg $\overline{AB}$

$$\vec{AB} = B - A = (1, -1, 3) \quad \vec{r}(t) = t(1, -1, 3) + (2, 0, 3) = (t+2, -t, 3t+3), \quad t \in [0, 1]$$

→ Hallar datos para calcular circulación

$$\vec{f}(\vec{r}(t)) = (t+2 + \vec{g}, -t + \vec{g}, 6t+6) \quad \vec{r}'(t) = (1, -1, 3)$$

→ calcular circulación

$$\begin{aligned} \int_0^1 (t+2 + \vec{g}, -t + \vec{g}, 6t+6) (1, -1, 3) dt &= \int_0^1 t+2 + \vec{g} + t - \vec{g} + 18t + 18 dt \\ &= \int_0^1 20t + 20 dt = 10t^2 + 20t \Big|_0^1 = 30 \end{aligned}$$